

Study of Nordgren's Equation of Hydraulic Fracturing

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Summary. A nonlinear partial differential equation modeling the propagation of a vertical hydraulic fracture first derived by Nordgren is studied. When properly posed, Nordgren's derivation constitutes a Stefan problem and requires another boundary condition—namely, that the velocity of the fluid at the crack tip equals the velocity of crack propagation. With this addition, Nordgren's similarity solution in the no-leakoff case is perfected by a power-series solution. New shut-in solutions are derived in the large- and no-leakoff cases where the crack of the large-leakoff shut-in solution retracts and the crack of the no-leakoff shut-in solution extends forever. This study ignores the effect of crack-tip rock strength on crack propagation.

Introduction

Nordgren¹ appears to be first to derive a partial differential equation modeling the dynamic propagation of a hydraulic fracture. He followed Perkins and Kern's² model, which couples an elastic equation relating pressure in a crack and crack width to a fluid equation relating flow to pressure and crack width. Nordgren, however, added the continuity equation of fluid flow relating flow to the rate of change of crack volume. This addition leads to a partial differential equation that he solved numerically. Nordgren's derivation includes boundary conditions but omits one that ordinarily is found in the specification of a Stefan problem³ where the salient feature is its moving boundary.

This study begins with Nordgren's nondimensional formulation. The moving-boundary condition is added and contrasted to a no-flow tip boundary condition. Nordgren's similarity solution method in the no-leakoff case is followed using the new boundary condition to split the problem into the solution of two ordinary differential equations. The new result is a crack-width profile that is determined analytically by a power series. Next, shut-in solutions in the large- and no-leakoff cases are derived. The paper concludes by showing why the Barenblatt tip condition fails to hold.

Nordgren's Equation

This study begins with Nordgren's nondimensional formulation, to which a fourth boundary condition has been added. Fig. 1 shows the fracture geometry of Nordgren's single-wing fracture. In Fig. 1, $w(x,t)$ denotes the width at the center of the fracture as a function of space x and time t . $L(t)$ is the crack length, $\tau(x)$ is the time it takes the fracture to reach point x , and h is the constant crack height.

Nordgren's equation (taken directly from his Eq. 18) is

$$(w^4)_{xx} = w_t + \frac{1}{\sqrt{t - \tau(x)}}, \quad 0 \leq x \leq L(t), \quad 0 \leq t, \quad \dots \quad (1)$$

with the following boundary conditions:

$$-(w^4)_x(0,t) = 1, \quad \dots \quad (2)$$

$$w(x,0) = 0, \quad \dots \quad (3)$$

$$w(L(t),t) = 0, \quad \dots \quad (4)$$

where $\tau(L(t)) = u$, and

$$\frac{-(w^4)_x(L(t),t)}{w(L(t),t)} = \frac{dL}{dt} \quad \dots \quad (5a)$$

$$\text{or } -(w^4)_x(L(t),t) = 0. \quad \dots \quad (5b)$$

Nordgren's equation and boundary conditions can be interpreted by expressing Eq. 1 in the form of a mass-balance law⁴:

$$(wv)_x w_t + \frac{1}{\sqrt{t - \tau(x)}} = 0,$$

where $v = -4w^2 w_x$ is the velocity of the fracturing fluid. Note that volume flow $q(x,t) = w(x,t)v(x,t) = -(w^4)_x$, so Eq. 1 says that the spatial rate of change of fluid volume flow is balanced by the time rate of increase in the volume of the fracture added to the volume of fluid leaking out of the fracture.

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The boundary condition in Eq. 2 states that the fluid volume flow into the fracture at the wellbore, $q(0,t)$, is constant. Eq. 3 states that the fracture width is zero at time zero; i.e., the crack starts to open at $t=0$. According to the boundary condition in Eq. 4, fracture width is zero at the crack tip. Here $L(t)$ is the inverse function of $\tau(x)$ as indicated in Eq. 4; more simply, $L(t)$ is the length of the fracture at time t . Note that the solution of the problem entails finding not only $w(x,t)$, but also $L(t)$ or $\tau(x)$. Two possibilities exist for the fourth boundary condition. Eq. 5a states that the velocity of the fracturing fluid at the crack tip equals the fracture propagation speed; i.e., $v(L(t),t) = dL/dt$. The alternative condition, Eq. 5b, states that the fluid volume at the tip, $q(L(t),t)$, is zero. Neither condition is set down in Nordgren's general statement of the crack problem. Eq. 5b, however, is stated and used to find the solution in the large-fluid-loss case—namely, when w_t is neglected in Eq. 1. Eq. 5a is not mentioned anywhere. Note that Eqs. 4 and 5a together imply Eq. 5b, but deriving Eq. 5a from 5b and the other boundary conditions is impossible. Eq. 5a, the Stefan condition,³ is always present in moving-boundary problems.

No-Leakoff Similarity Solution

Eq. 1 without leakoff can be solved analytically by the similarity transformation $w(x,t) = T(t)\psi(\xi)$, where $\xi = x/L(t)$. This transformation says that all crack profiles are similar to $\psi(\xi)$ where the scale factor at time t is $T(t)$. The boundary condition in Eq. 2 transforms to

$$-(w^4)_x(0,t) = 1 = [-T^4(t)/L(t)](d\psi^4/d\xi)(0),$$

which implies $T^4(t) = L(t)$ and $(d\psi^4/d\xi)(0) = -1$. $w^4(x,0) = 0 = L(0)\psi^4(\xi)$ implies $L(0) = 0$, and $w^4(L(t),t) = 0 = L(t)\psi^4(1)$ implies $\psi(1) = 0$.

Substituting $w(x,t) = L^{1/4}(t)\psi(\xi)$ into Eq. 1 separates Eq. 1 into two ordinary differential equations. One is a function of t alone and the other a function of ξ alone:

$$5/(dL^{3/4}/dt)(t) = [\psi(\xi) - 4\xi(d\psi/d\xi)]/(d^2\psi^4/d\xi^2) = \lambda. \quad \dots \quad (6)$$

Integrating the first part of Eq. 6 gives $L(t) = (\text{constant} + 5t/\lambda)^{4/5}$, but $L(0) = 0$, so $L(t) = (5t/\lambda)^{4/5}$, as obtained by Nordgren. Substituting the similarity form in Eq. 5a and using the relation between L and λ as given by Eq. 6 leads to the following condition on ψ :

$$(1/\psi)(d\psi^4/d\xi)(1) = -4/\lambda,$$

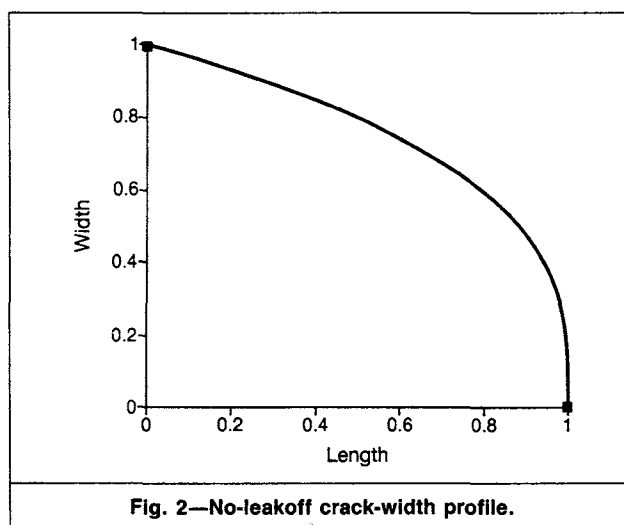
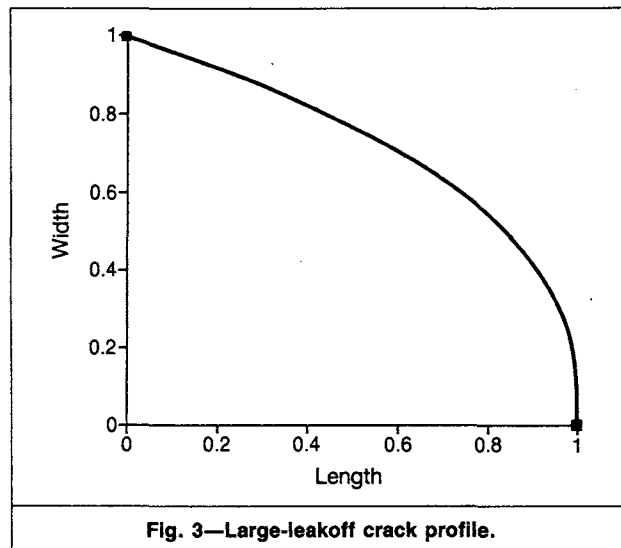
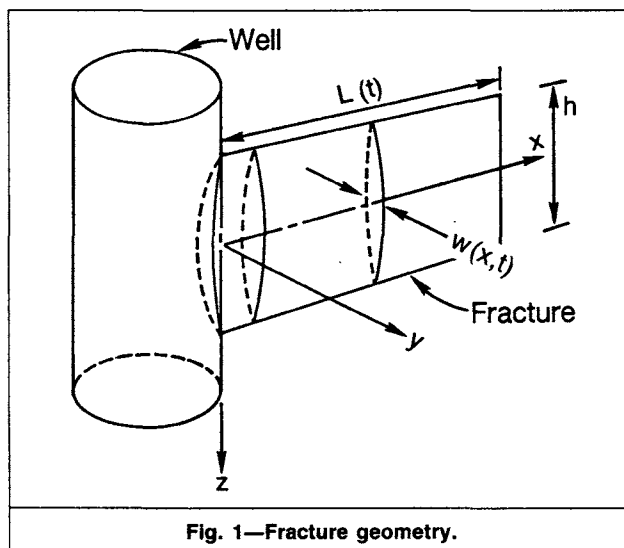
where the condition is understood as a limit as $\xi \rightarrow 1$. The second part of Eq. 6 is a second-degree ordinary differential equation in the independent variable ξ involving the free parameter λ . Three conditions needed to solve it uniquely are $(d\psi^4/d\xi)(0) = -1$, $\psi(1) = 0$, and $(1/\psi)(d\psi^4/d\xi)(1) = -4/\lambda$.

Assume that $\psi^4(\xi) = (1-\xi)^\alpha f(1-\xi)$, where $\alpha > 0$ and $f(1-\xi)$ is analytic (i.e., has a power series) and is nonzero at $\xi = 1$. The standard procedure of substitution and equating like powers leads to

$$\psi^4(\xi) = (1-\xi)^{3/5} (3/\lambda)^{3/5}$$

$$\times [1 - (1/24)(1-\xi) + (67/32,257)(1-\xi)^2 - (7/105)(1-\xi)^3 + \dots],$$

where $\lambda = 3.53106$, so $w(x,t) = (5t/\lambda)^{1/5} \psi(\xi)$.



The no-leakoff crack profile $w(x,t)/w(0,t)$ in Fig. 2 appears identical to that of the large-leakoff solution in Fig. 3. Overlaying the graphs, however, reveals that the large-leakoff profile, which can be shown to behave like $(1-\xi)^{3/2}$ at the tip, falls below the no-leakoff profile. This has been verified analytically and numerically and is at variance with Nordgren's Fig. 4, which shows the two profiles crossing at $\xi=0.9$. Nordgren's numerical finite-difference solution used to produce the no-leakoff similarity solution apparently is not accurate near the crack tip, failing to produce the proper $(1-\xi)^{3/2}$ behavior there.

Large-Leakoff Shut-In Solution

The large-leakoff case ignores the w_t term in Eq. 1, so Eq. 1 is simply

$$-q_x = 1/\sqrt{t-\tau(x)} \quad (7)$$

Integrating Eq. 7 and using Eq. 5b leads to

$$q(0,t) = -q \Big|_{x=0}^{x=L(t)} = \int_0^{L(t)} \frac{dx}{\sqrt{t-\tau(x)}} = \int_0^{L(t)} \frac{L'(u) du}{\sqrt{t-u}} \quad (8)$$

But now we take $q(0,t) = 1-H(t-t_o)$, where $H(t)$ is the Heaviside unit step function. Pumping stops at time t_o , the well is shut in, and the flow at the wellbore goes to zero. Substitution of this form for $q(0,t)$ into Eq. 8 gives

$$1-H(t-t_o) = \int_0^{L(t)} \frac{L'(u) du}{\sqrt{t-u}} \quad (9)$$

Application of the Laplace transform yields

$$\frac{1}{s} - \frac{1}{s} e^{-st_o} = \mathcal{L}(L') \mathcal{L}\left(\frac{1}{\sqrt{t}}\right) = [s\mathcal{L}(L) - L(0)] \sqrt{\frac{\pi}{s}}$$

Because $L(0)=0$,

$$\mathcal{L}(L) = \frac{1}{s^2} \frac{1-e^{-st_o}}{\sqrt{\pi/s}} = \frac{s^{-3/2}}{\sqrt{\pi}} - \frac{s^{-3/2}}{\sqrt{\pi}} e^{-st_o},$$

which, upon inversion, yields

$$L(t) = (2/\pi)\sqrt{t} - (2/\pi)\sqrt{t-t_o} H(t-t_o) \quad (10)$$

Eq. 10 says that the crack length shortens to zero after pumping stops, so the crack closes up and yields an inverse function $\tau(x)$ for the time it takes the crack to open to distance x :

$$\tau(x) = [(\pi/2)x]^2, \quad x \leq (2/\pi)\sqrt{t_o} \quad (t \leq t_o),$$

$$\text{and } \tau(x) = \frac{1}{4} \left(\frac{\pi}{2} x \right)^2 + \frac{1}{4} \frac{t_o^2}{[(\pi/2)x]^2} + \frac{t_o}{2}, \quad x \leq \frac{2}{\pi} \sqrt{t_o} \quad (t > t_o).$$

The first part of this expression is the nondimensional form of Nordgren's Eq. B-5 for the large-leakoff solution. The second part is understood to be evaluated with crack lengths x tending to zero from the maximum length of $(2/\pi)\sqrt{t_o}$ attained at t_o . In fact, this expression is the time at which the crack closes at position x .

Until time t_o , crack behavior is identical to that of the large-leakoff solution. After t_o , integration of Eq. 7 from x to $L(t) = (2/\pi)(\sqrt{t} - \sqrt{t-t_o})$ gives

$$\begin{aligned} q(x,t) &= -q(L(t),t) + q(x,t) = \int_x^{L(t)} \frac{du}{\sqrt{t-\tau(u)}} \\ &= \int_x^{L(t)} \frac{du}{\sqrt{t-[(\pi/2)u]^2}} \\ &= \frac{2}{\pi} \left[\sin^{-1} \frac{L(t)}{(2/\pi)\sqrt{t}} - \sin^{-1} \frac{x}{(2/\pi)\sqrt{t}} \right], \\ 0 &\leq x \leq L(t), \quad t_o \leq t. \end{aligned} \quad (11)$$

Note that the boundary condition $q(0,t)=0$ for $t \geq t_o$ is not satisfied, but $q(0,t) \rightarrow 0^+$ as $t \rightarrow \infty$; however, the boundary condition $q(L(t),t)=0$ is satisfied.

Integration of Eq. 11 from x to $L(t)$, where $L_o(t) = (2/\pi)\sqrt{t}$, gives

$$-(w^4) \Big|_x^{L(t)} = w^4(x,t) = \int_x^{L(t)} \frac{2}{\pi} \left[\sin^{-1} \frac{L(t)}{L_o(t)} - \sin^{-1} \frac{u}{L_o(t)} \right] du$$

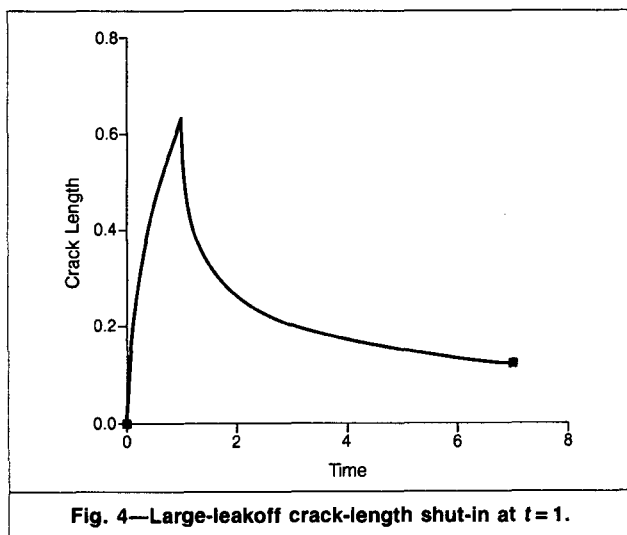


Fig. 4—Large-leakoff crack-length shut-in at $t = 1$.

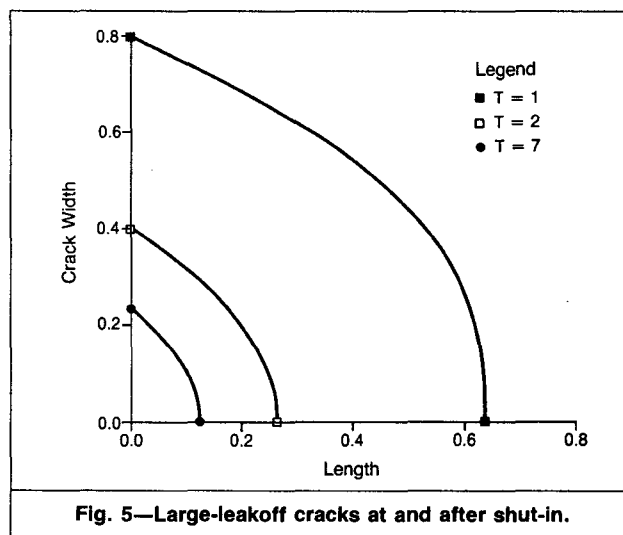


Fig. 5—Large-leakoff cracks at and after shut-in.

$$= L_o(t) \frac{2}{\pi} \left\{ \frac{x}{L_o(t)} \left[\sin^{-1} \frac{x}{L_o(t)} - \sin^{-1} \frac{L(t)}{L_o(t)} \right] + \sqrt{1 - \left[\frac{x}{L_o(t)} \right]^2} - \sqrt{1 - \left[\frac{L(t)}{L_o(t)} \right]^2} \right\}, \dots (12)$$

which is valid for $0 \leq x \leq L(t)$, $t_o \leq t$. Substituting $L(t) = L_o(t)$ gives Nordgren's large-leakoff solution (his Eq. B-8). Fig. 4 shows crack length as given by Eq. 10 before and after shut-in at $t_o = 1$. Fig. 5 displays $w(x, t)/w_o(t)$ as determined by Eq. 12.

No-Leakoff Shut-In Solution

When pumping stops at t_o and the well is shut in, crack width is given by the no-leakoff similarity solution. The time origin is moved to t_o , so crack width becomes an initial condition and the problem is identical to that specified by Eqs. 1 through 5a with the following changes. The leakoff term is missing in Eq. 1 and the boundary conditions in Eqs. 2 and 3 become $-(w^4)_x(0, t) = 0$ and $w(x, 0) = w_o(x)$, respectively. $w_o(x)$ is the initial crack width determined from the similarity solution at $t = t_o$, which shows that the crack has extended a finite amount, say $a = L(t_o)$, and that $w_o(x)$ has support on the interval $0 \leq x \leq a$.

Gurtin *et al.*⁴ addressed this problem and offered a solution algorithm for accurate tracking of the free boundary $L(t)$. The method uses a transformation from the spatial coordinate x to a material coordinate p :

$$x = X(p, t), \quad X(p, 0) = p, \quad 0 \leq p \leq a.$$

Thus, $x = X(p, t)$ represents the motion of the crack with points on the crack labeled by their positions p at $t = 0$. Note that at the free boundary $L(t) = X(a, t)$. Gurtin *et al.* showed that by defining $U(p, t) = w(X(p, t), t)$, the problem as stated transforms to the following initial value problem in X with intermediate variable U and independent variables p and t . Solve

$$X_t(p, t) = -4w_o^2(p)U_p(p, t)/X_p^3(p, t) \dots (13)$$

$$\text{and } U(p, t) = w_o(p)/X_p(p, t) \dots (14)$$

subject to $X(p, 0) = p$, $X(0, t) = 0$, $U_p(0, t) = 0$, $0 \leq p \leq a$.

Gurtin *et al.* suggested the following partially explicit finite-difference scheme. Choose $\Delta t > 0$ and $\Delta p = a/N$ for some integer N , introduce the mesh

$$p_n = n\Delta p, \quad n = 0, 1, 2, \dots, N,$$

$$\text{and } t_i = i\Delta t, \quad i = 0, 1, 2, \dots,$$

and write f_n^i for the value of a function f at the mesh point (p_n, t_i) . Then Eqs. 13 and 14 are approximated by

$$X_n^{i+1} = X_n^i - [\Delta t 4w_n^2 L_n U^i / (M_n X_n^i)^3] \dots (15)$$

$$\text{and } U_n^{i+1} = w_n / M_n X_n^{i+1}. \dots (16)$$

In Eqs. 15 and 16, $w_n = w_o(p_n)$ and the spatial difference operators are

$$L_n f = \frac{1}{2\Delta p} \begin{cases} 0 & n=0 \\ -f_{n-1} + f_{n+1} & 1 \leq n \leq N-1 \\ f_{N-2} - 4f_{N-1} & n=N \end{cases}$$

$$\text{and } M_n f = \frac{1}{2\Delta p} \begin{cases} 2f_1 & n=0 \\ -f_{n-1} + f_{n+1} & 1 \leq n \leq N-1 \\ f_{N-2} - 4f_{N-1} + 3f_N & n=N \end{cases}$$

When $n = N$, Eq. 15 yields no change in X_N from one timestep to the next because $w_N = w_o(a) = 0$; thus, according to Eq. 15, the crack will never extend. To derive the correct condition, one sees that, after substituting Eq. 14 into Eq. 13,

$$X_t(p, t) = \frac{-4w_o^2(p)}{X_p^3} \left[\frac{w_o'(p)X_p - w_o(p)X_{pp}}{X_p^2} \right],$$

which, when evaluated at a , gives

$$X_t(a, t) = [-4w_o^2(a)w_o'(a)X_p(a, t) + 4w_o^3(a)X_{pp}(a, t)]/X_p^5(a, t). \dots (17)$$

Now $w_o(a, t) = 0$, so at first glance it again appears that $X_t(a, t) = 0$; however, this is not the case because the equation is to be understood as a limit as $p \rightarrow a$. With this proviso,

$$\lim_{p \rightarrow a} -4w_o^2(p)w_o'(p) = \frac{dL}{dt} \bigg|_{t=t_o} = \frac{d}{dt} (5t/\lambda)^{1/5} \bigg|_{t=t_o} = \frac{4}{\lambda} L(t_o)^{-4/5} = \frac{4}{a^{4/5} \lambda},$$

where λ is the similarity constant 3.53106 obtained earlier. The limit is v , the speed of the crack of the no-leakoff similarity solution when pumping is stopped at t_o . Therefore, Eq. 17 is taken as

$$X_t(a, t) = 4/[a^{4/5} \lambda X_p^4(a, t)]$$

and is approximated by

$$X_N^{i+1} = X_N^i + 4\Delta t/[a^{4/5} \lambda (M_N X_N^i)^4].$$

This replaces Eq. 15 when $n = N$ and shows that the crack always extends after shut-in.

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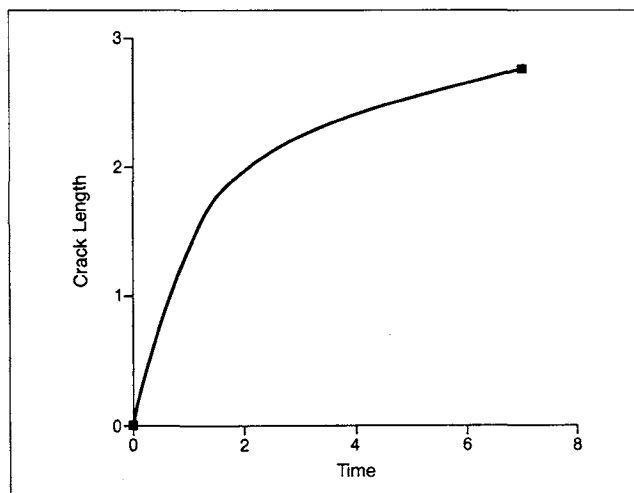


Fig. 6—No-leakoff crack-length shut-in at $t=1$.

Gurtin *et al.* remarked that the p =constant trajectories $X(p,t)$ are analogs of characteristic curves as the degeneracy of Eq. 1 propagates along such trajectories. Indeed, Eq. 1 degenerates at the crack tip because derivatives are discontinuous there and $w(X(t,p),t)=0$ if and only if $w_o(p)=0$.

Figs. 6 and 7 are the result of a FORTRAN program implementation of Eqs. 15 and 16 and show that the crack extends forever, but ever more slowly, becoming narrower as it approaches the steady-state solution of Eq. 1, which is $w(x,0)=0$.

Conclusions

This study improves the mathematics of Nordgren's development and solution of his hydraulic fracturing model and extends the model to shut-in cases. The no-leakoff similarity solution shows that crack-tip width behaves as the one-third power of the distance from the crack tip. Therefore, the Barenblatt condition^{5,6} that the tip close smoothly [i.e., $\psi_\xi(1)=0$] is not satisfied.

Nomenclature

- $a = L(t_o)$, dummy variable for fracture length at shut-in
- h = constant crack height
- $H(t)$ = Heaviside unit step function
- $\mathcal{L}$ = Laplace transform operator
- L_n, M_n = finite-difference spatial operators
- $L(t)$ = fracture length at time t
- $L_o(t)$ = fracture length before shut-in
- p = material coordinate locating positions along crack at shut-in
- $\Delta p = p$ increment, a/N
- $q(x,t)$ = fluid volume flow rate
- s = Laplace transform variable

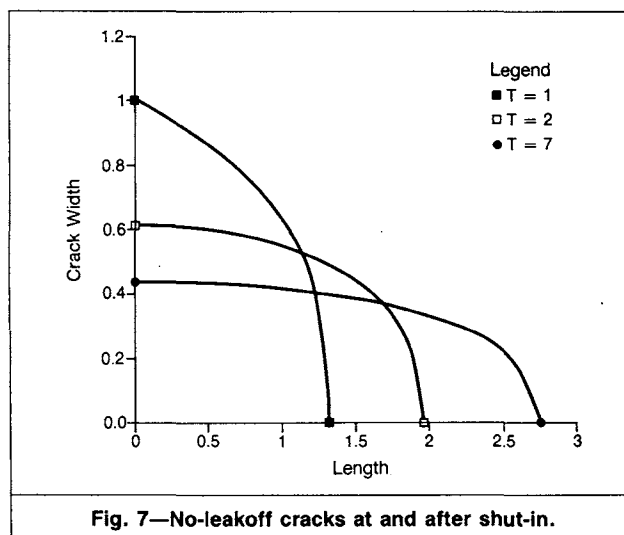


Fig. 7—No-leakoff cracks at and after shut-in.

- t = time
- t_i = i th time position ($i\Delta t$)
- Δt = time increment
- $T(t)$ = time-similarity separation function
- u = dummy variable of integration
- $U(p,t)$ = intermediate variable $w(X(p,t),t)$
- v = fracturing-fluid velocity
- w = crack width
- $w(x,t)$ = maximum crack width
- $w_o(x)$ = fracture width at shut-in time t_o
- x = coordinate in fracture direction
- Δx = x increment
- $X(p,t)$ = coordinate along fracture after shut-in
- α = fractional power in series expansion
- λ = constant in separation of variables
- ξ = spatial similarity coordinate, $x/L(t)$
- $\tau(x)$ = time for fracture to reach x
- $\psi(\xi)$ = spatial similarity separation function

Subscripts

- n = index for material coordinate p
- N = number of Δp increments

Superscript

- i = time index

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